

Muhammad Zubair Zafar

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Founding engineer shipping LLM agents and applied ML for health and longevity. Full-stack (Python, TypeScript), zero-to-one product owner.

EXPERIENCE

- Andever.ai** | Founding Engineer (first engineering hire) San Francisco, CA | Jun 2026 – Present
- Sole engineer at a stealth longevity-AI startup; scoping v1 with the founder and investors across 4 target segments (private clinics, pro-sports medical teams, aging-diagnostics hardware, insurers) and owning architecture, build, deployment, and iteration.
 - Designing and building a 3-agent platform (analysis, routine/calendar, fitness) in Python and TypeScript that turns wearable, lab, and biomarker data into 2 headline metrics, biological age and projected lifespan, using open-source aging models, prompt engineering, and an eval set rather than proprietary training.
 - Integrating a third-party health-data platform over REST with consent-first storage, Amazon supplement purchasing, and calendar scheduling; deploying on-site with design-partner clinics in a forward-deployed engineering model.
- Eastern Illinois University** | Software Engineer Charleston, IL | Aug 2024 – May 2026
- Architected and shipped an enterprise HR system for 7,000+ users with role-based auth; cut API response time 65% and kept it stable through peak enrollment periods.
 - Automated HR workflows via Twilio-integrated REST APIs, cutting manual data entry 40% and request turnaround 25%.
 - Containerized services with Docker, deployed via CI/CD with campus sysadmins, and owned the production SQL data layer.
- WPBrigade** | Backend Software Engineer Sialkot, Pakistan (remote) | Dec 2023 – Aug 2024
- Built a modular site-building framework (PHP, MySQL, JS) used by 200+ non-technical users, cutting deployment time 60%; mentored 3 junior engineers on architecture and secure coding.
- Etihad Aviation Group** | Software Engineering Intern Abu Dhabi, UAE | Sep 2023 – Oct 2023
- Productionized a hackathon-winning cargo-message parser (Python, Flask) that auto-corrects performance messages, cutting manual processing 30% and eliminating 95% of data-entry errors.
 - Built a real-time visa/passport requirements service integrating the Timatic REST API with spaCy NLP for 200+ countries.

RESEARCH

- Eastern Illinois University** | Graduate Research Assistant, Medical Imaging ML Charleston, IL | Aug 2025 – May 2026
- Trained YOLOv8/PyTorch models for multi-class classification of 80,000+ labeled MRI scans: 97.2% test accuracy, 95.8% F1.
 - Built the full pipeline (preprocessing, augmentation, GPU training, automated eval with early stopping), cutting training time 70%.

SELECTED PROJECTS

21 projects at zubair480.github.io/projects

- EstateAgent AI** | Hyper-local real-estate intelligence, HackIllinois 2026 | [GitHub](#) | [Live demo](#) Mar 2026
- Multi-agent LLM app (Qwen 2.5, Llama 3.1) with FastAPI on Modal serverless GPUs and a Cloudflare edge layer (Workers, Durable Objects, KV, D1, R2, Vectorize) streaming pipeline progress over stateful WebSockets.
 - Runs 5,000 parallel Monte Carlo simulations in under 12 seconds by bridging Modal GPUs with Cloudflare edge workers.
- Attest** | 2nd place (solo), Agent Native Builders Hackathon at Cloudflare SF HQ | [GitHub](#) 2026
- Replaced static vendor-compliance questionnaires with a bank agent that interrogates a vendor agent for proof; a two-stage evaluation gate decides what counts as evidence (multi-agent, Cotal.ai, Runtype.ai, JavaScript).
- ROOT** | 2nd place, CrewAI Hackathon | [Project post](#) 2026
- 14 coordinated agents read a Zendesk onboarding ticket and provision the software, hardware, and access a new hire needs, turning a week of manual provisioning into minutes (CrewAI, Zendesk API, full stack).
- Clip Police** | 2nd place, Auth0 × Stripe Hackathon at Okta SF HQ | [Project post](#) 2026
- Multi-agent platform that finds unauthorized reposts of creators' videos across the web and assembles a real-time circulation report, with Auth0 identity and Stripe billing.
- crampon** | Reinforcement learning, Himalaya Robotics Hack | [GitHub](#) 2026
- Trained a MuJoCo locomotion policy that keeps a Unitree G1 humanoid upright on simulated Himalayan ice, modeling ice friction, cold actuators, and high-altitude wind; 2× to 6.6× longer survival with no regression on flat ground.

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Java, C++

AI / ML: LLM agent orchestration (CrewAI, MCP), prompt engineering, evals, RAG, OpenAI & Anthropic APIs, Llama, Qwen, PyTorch, reinforcement learning (MuJoCo), YOLOv8, OpenCV

Web & Cloud: FastAPI, Flask, Node.js, React, REST & WebSocket APIs, PostgreSQL, MySQL, MongoDB, Docker, CI/CD, Cloudflare Workers & Durable Objects, Modal, MCP

EDUCATION

Eastern Illinois University | M.S. Computer Information Technology, GPA 4.0

Aug 2024 – May 2026

Government College University Faisalabad | B.S. Computer Science

Oct 2019 – Sep 2023